01/07/2026

CODE-B

Aakash
Medical | IIT-JEE | Foundations

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FORTNIGHTLY TEST FOR NEET-2027 (XII PASSED) PH-2

MM: 720 **FT-01** **Time: 180 Mins.**

Answers

1. (3)	37. (4)	73. (1)	109. (1)	145. (3)
2. (1)	38. (2)	74. (1)	110. (1)	146. (3)
3. (4)	39. (4)	75. (1)	111. (3)	147. (2)
4. (4)	40. (2)	76. (3)	112. (2)	148. (1)
5. (2)	41. (1)	77. (3)	113. (4)	149. (1)
6. (2)	42. (2)	78. (1)	114. (4)	150. (3)
7. (1)	43. (3)	79. (3)	115. (4)	151. (2)
8. (1)	44. (3)	80. (2)	116. (2)	152. (3)
9. (4)	45. (1)	81. (3)	117. (3)	153. (1)
10. (1)	46. (3)	82. (2)	118. (2)	154. (1)
11. (2)	47. (3)	83. (1)	119. (2)	155. (4)
12. (2)	48. (2)	84. (4)	120. (2)	156. (4)
13. (1)	49. (3)	85. (3)	121. (4)	157. (1)
14. (2)	50. (3)	86. (2)	122. (2)	158. (3)
15. (4)	51. (2)	87. (2)	123. (3)	159. (2)
16. (1)	52. (4)	88. (2)	124. (4)	160. (1)
17. (4)	53. (4)	89. (4)	125. (3)	161. (3)
18. (1)	54. (3)	90. (3)	126. (3)	162. (2)
19. (1)	55. (2)	91. (3)	127. (3)	163. (3)
20. (2)	56. (4)	92. (3)	128. (2)	164. (4)
21. (4)	57. (4)	93. (2)	129. (3)	165. (2)
22. (3)	58. (1)	94. (2)	130. (1)	166. (3)
23. (2)	59. (1)	95. (3)	131. (2)	167. (2)
24. (2)	60. (3)	96. (3)	132. (1)	168. (3)
25. (3)	61. (4)	97. (2)	133. (2)	169. (4)
26. (4)	62. (1)	98. (1)	134. (3)	170. (3)
27. (1)	63. (3)	99. (2)	135. (1)	171. (1)
28. (4)	64. (2)	100. (4)	136. (1)	172. (4)
29. (4)	65. (3)	101. (3)	137. (4)	173. (2)
30. (2)	66. (2)	102. (3)	138. (1)	174. (1)
31. (4)	67. (3)	103. (4)	139. (4)	175. (2)
32. (1)	68. (4)	104. (4)	140. (4)	176. (2)
33. (1)	69. (2)	105. (3)	141. (4)	177. (4)
34. (3)	70. (3)	106. (4)	142. (2)	178. (1)
35. (2)	71. (3)	107. (3)	143. (4)	179. (4)
36. (4)	72. (1)	108. (2)	144. (4)	180. (2)

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Hints and Solutions

PHYSICS

(1) **Answer :** (3)
Solution:
$R_{eq} = R_1 + R_2 = 5 + 10 = 15\Omega$
$\Delta R_{eq} = \Delta R_1 + \Delta R_2 = 0.2 + 0.1 = 0.3$
$\frac{\Delta R_{eq}}{R_{eq}} = \frac{0.3}{15} \times 100 = 2\%$

(2) **Answer :** (1)
Solution:
$KE = \frac{P^2}{2m}$
$\frac{\Delta KE}{KE} \times 100 = 2\left(\frac{\Delta P}{P} \times 100\right)$
$\frac{\Delta KE}{KE} \times 100\% = 2 \times 1.5\% = 3\%$

(3) **Answer :** (4)
Hint:
Using rules of significant figures.
Solution:
$\frac{62.24}{2.0} = 31.12$
= 31 (two significant figures)

(4) **Answer :** (4)
Solution:
1.001 m is closest to 1 m

(5) **Answer :** (2)
Solution:
a = 1.2 m
$A = a^2 = 1.2 \times 1.2 = 1.44\ m^2$
$A \approx 1.4\ m^2$

(6) **Answer :** (2)
Solution:
Dimensional correctness of a relation does not ensure its physical correctness.

(7) **Answer :** (1)
Hint:
Use rules of round off.
Solution:

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17.684
→ Uncertain digit $(>5) \Rightarrow 17.7$

17.65
→ Preceding digit is even $\Rightarrow 17.6$

17.75
→ Preceding digit is odd $\Rightarrow 17.8$

17.632
→ Uncertain digit $(<5) \Rightarrow 17.6$

(8) **Answer :** (1)
Solution:
9.99 – 0.009 = 9.981 should be reported to least no of decimal places so answer should be 9.98 m.

(9) **Answer :** (4)
Solution:
Angular momentum, $L = \frac{nh}{2\pi}$
∴ Unit of angular momentum (*L*) is same as Planck's constant.
E = *hf* ⇒ $h = \frac{E}{f}$
∴ Unit of $\frac{\text{Energy}}{\text{Frequency}}$ is same as Planck's constant.

(10) **Answer :** (1)
Solution:
True value = Mean value $= \frac{10+11+10+10+9}{5}$ =10 cm

(11) **Answer :** (2)
Solution:
By theory

(12) **Answer :** (2)
Solution:
Maximum % error in $P = \frac{1}{2}[2+2+(4\times 2)]+3$
= 9%

(13) **Answer :** (1)
Hint:
Use principle of homogeneity.
Solution:
$[a] = [pt] = \left[\frac{q}{t^2+r}\right]$
$[p] = \left[\frac{a}{t}\right]$ and $[q] = [at^2]$ and $[r] = [t^2]$
$\Rightarrow \left[\frac{pr}{q}\right] = \left[\frac{\frac{a}{t}\times t^2}{at^2}\right] = \left[\frac{1}{t}\right] = [T^{-1}]$

(14) **Answer :** (2)
Hint:
$\frac{\text{Pitch}}{\text{Number of division on circular scale}} = \text{Least count}$
Solution:
Pitch = 50 × 0.01 = 0.5 mm

(15) **Answer :** (4)

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Hint:
Principle of Homogeneity
Solution:
If *A* and *B* have same dimensions then they can be either added or subtracted while *C* can be either divided or multiplied with them.

(16) **Answer :** (1)
Solution:
Velocity → $[LT^{-1}]$
Acceleration → $[LT^{-2}]$
Force → $[MLT^{-2}]$
Pressure → $[ML^{-1}T^{-2}]$

(17) **Answer :** (4)
Hint:
Time constant (τ) = *RC*
Solution:
[*Q*] = [current] [time]
[current] = $\left[\frac{Q}{\text{time}}\right] = \left[\frac{Q}{RC}\right]$

(18) **Answer :** (1)
Solution:
Reading = MSR + (CSR)LC
= 6 + 51(0.01)
= 6.51 mm

(19) **Answer :** (1)
Solution:
Least Count = 1 MSD – 1 VSD
= 2 $\left(1-\frac{9}{10}\right)$ = 0.2 mm
Zero error = – (10 – 5) × 0.2 mm
= –1.0 mm

(20) **Answer :** (2)
Hint:
Reading = MSR + (L.C.) × VSD
Solution:
L.C. = 1 MSD – 1 VSD
$= 1-\frac{18}{20}$MSD
$= \frac{1}{10}$mm
Reading $= 25+5\times\frac{1}{10}$ = 25.5 mm

(21) **Answer :** (4)
Solution:
L.C = 1 MSD –1VSD
& 5 VSD = 4 MSD
∴L.C = $\left(1-\frac{4}{5}\right)\frac{1\,\text{cm}}{8}$
$L.C = \left(\frac{1}{5}\times\frac{1}{8}\right)\text{cm} = \frac{1}{4}\text{ mm}$

(22) **Answer :** (3)
Solution:
Since acceleration is constant, hence its average value is equal to *g*.
Average speed < v > = $\frac{\text{Total distance}}{\text{Total time}} = \frac{2H}{T}$
$< v >= \frac{2\times\frac{v^2}{2g}}{\frac{2v}{g}} = \frac{v}{2}$

(23) **Answer :** (2)
Solution:

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Given, $t_2 - t_1 = 10$ s

$t_2 + t_1 = \frac{2u}{g} = \frac{2\times 52}{10} = 10.4$

$\Rightarrow 2t_2 = 20.4$

$\Rightarrow \ t_2 = 10.2$ s

$t_1 = 0.2$ s

So, $t_1 t_2 = \frac{2h}{g}$

$0.2 \times 10.2 = \frac{2\times h}{10}$

$\Rightarrow 1 \times 10.2 = h \Rightarrow \boxed{10.2m = h}$

(24) Answer : (2)
Solution:

$S_{2.5} - S_2 = ?$ (distance travelled in 5th half second)

$S_{2.5} = ut + \frac{1}{2}at^2$

$\Rightarrow \ S_{2.5} = \frac{1}{2} \times 2 \times (2.5)^2 = 6.25$ m ($\because$ u = 0)

$S_2 = \frac{1}{2} \times 2 \times 4 = 4$ m

So, $\boxed{S_{2.5} - S_2 = 2.25 \text{ m}}$

(25) Answer : (3)
Solution:

$t_{\text{meet}} = \frac{h}{u} = \frac{100}{50} = 2s$

Required = $ut - \frac{1}{2}gt^2 = 50 \times 2 - \frac{1}{2} \times 10 \times 2^2 \quad = 100 - 20 = 80$ m

(26) Answer : (4)
Solution:

Average acceleration $\langle a \rangle = \frac{\text{Final velocity – Initial velocity}}{\text{Total time}}$

$\therefore \langle a \rangle = 0$

Total displacement = $\left(\frac{1}{2} \times 2 \times 4\right) + [(6-2) \times 4] + \left(\frac{1}{2} \times 2 \times 4\right) = 24$m

Average velocity $< v > \ = \ \frac{\text{Displacement}}{\text{Time}}$

$< v > \ = \ \frac{24}{8} = 3$ m/s

(27) Answer : (1)
Solution:

$(v^2 - 0) = -4(t - 2)$

$v^2 = 8 - 4t$

At $t = 1$ s, $v^2 = 8 - 4$

$v = \pm 2$ m/s

$2v\frac{dv}{dt} = 0 - 4$

$2 \times \left(\pm 2\right) \frac{dv}{dt} = -4$

$\frac{dv}{dt} = \pm 1$ m/s^2

$\left|\frac{dv}{dt}\right| \ = \ 1$ m/s^2

(28) Answer : (4)
Solution:

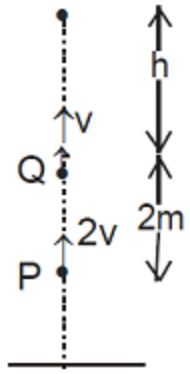
$x^2 = 2t^2 + 6t$

Differentiating 2x $2x\frac{dx}{dt} = 4t + 6$

$v = \frac{2(2t+3)}{2x} = \frac{2t+3}{x}$

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(29) **Answer :** (4)
Solution:
$v^2 - u^2 = 2as$



For part PQ, $v^2 - (2v)^2 = 2(-g)2$
$-3v^2 = -4g$
$v^2 = \frac{4g}{3} \Rightarrow v = \sqrt{\frac{4g}{3}}$
If h is further height, then
$O^2 - \left(\sqrt{\frac{4g}{3}}\right)^2 = 2(-g)h$
$-\frac{4g}{3} = -2gh \Rightarrow h = \frac{2}{3}m$

(30) **Answer :** (2)
Solution:
For uniformly accelerated motion in straight line
$v_{avg} = \frac{u+v}{2}$
$v_{avg} = \frac{0+(0+at)}{2} = \frac{at}{2}$

(31) **Answer :** (4)
Hint:
$t_a = \frac{U}{(g+a)}$
Where *a* is retardation due to air drag only.
Solution:
When body is moving upward air friction is acting downward. So
$t_a = \frac{U}{(g+a)}$;
where *a* is retardation due to air friction only.
$\Rightarrow\ 1 = \frac{15}{(10+a)} \Rightarrow a = 15 - 10 = 5\ \text{m/s}^2$
Height attained $H = \frac{1}{2}(15) \times 1 = 7.5$ m.
For downward journey, air friction acts upwards. So time of descent
$t_d = \sqrt{\frac{2H}{(g-a)}} = \sqrt{\frac{15}{(10-5)}} = \sqrt{3} = 1.7$ s

(32) **Answer :** (1)
Solution:
During *t* = 0 to *t* = 4 s, the slope of *x*-*t* graph is zero four times means the particle is at rest four times.

(33) **Answer :** (1)
Solution:
$v = kx^{3/2}$
$a = v\frac{dv}{dx}$, $\frac{dv}{dx} = k \cdot \frac{3}{2}x^{1/2}$
$a = \frac{3}{2}k^2x^2$
∴ Graph is parabola

(34) **Answer :** (3)
Solution:

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v_A = tan 37°, v_B = –tan 37°

= $\frac{3}{4}$ $= -\frac{3}{4}$

$v_{AB} = v_A - v_B = \frac{3}{2}$ m/s

(35) **Answer :** (2)
Solution:
Area under the *a* – *t* curve should be zero.

(36) **Answer :** (4)
Solution:

$V_{av} = \frac{3S}{\frac{S}{60}+\frac{S}{20}+\frac{S}{10}}$ = 18 kmph

(37) **Answer :** (4)
Solution:
Acceleration is positive in all three cases.

(38) **Answer :** (2)
Solution:
$a = \frac{dv}{dt} = 2t-1$

Acceleration is positive in interval $\frac{1}{2} < t < 1$ and velocity is negative

(39) **Answer :** (4)
Solution:
In this case, the minimum velocity of bullet to hit the car must be $\sqrt{2ad}$
⇒ $v = \sqrt{2 \times 20 \times 2000}$
$v = \sqrt{8 \times 10^4} = 2\sqrt{2} \times 100$ m/s
but, velocity is 200 m/s hence, it will not hit thief's car.

(40) **Answer :** (2)
Solution:
Area under velocity-time graph gives displacement and in one dimensional motion, constant speed in an interval have zero acceleration.

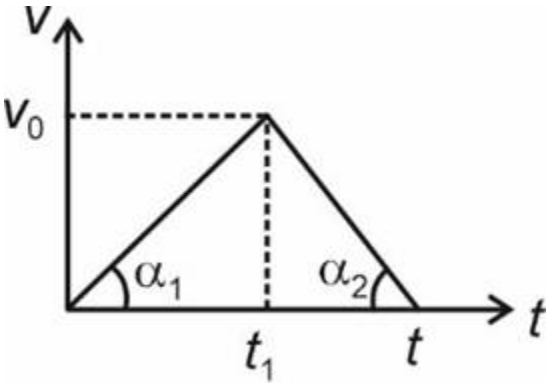
(41) **Answer :** (1)
Solution:
The zero velocity of a particle at any instant does not necessarily imply zero acceleration at that instant.
⇒ | Displacement | ≤ Distance
⇒ For a uniform motion, velocity is constant.
⇒ For uniform accelerated motion,
$x = ut + \frac{1}{2}at^2$
Position-time graph is a parabola.

(42) **Answer :** (2)
Solution:

Change in velocity	– Area under acceleration vs time curve
Acceleration	– Slope of velocity vs time curve
Displacement	– Area under velocity vs time curve
Instantaneous velocity	– Slope of position vs time curve

(43) **Answer :** (3)
Solution:
Make velocity time graph of given condition,

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$\tan\alpha_1 = \frac{v_0}{t_1} = a_1$ ...(i)

$\tan\alpha_2 = \frac{v_0}{t-t_1} = a_2 \ \frac{v_0}{a_2} + t_1 = t$

$\Rightarrow \ \frac{v_0}{a_2} + \frac{v_0}{a_1} = t$

[From equation (i)]

$\Rightarrow \ v_0\left(\frac{a_1+a_2}{a_1a_2}\right) = t$

$\Rightarrow \ v_0 = \frac{ta_1a_2}{a_1+a_2}$

(44) Answer : (3)
Solution:
Displacement is given by area under velocity-time graph
Displacement covered by I is more than II is not true at all times.

(45) Answer : (1)
Solution:
Acceleration of bolt with respect to elevator when bolt starts falling is
$a_{BE} = 10 - (-2) \ = 12\ \text{m/s}^2$ downward
Initial relative velocity = 0
Displacement travelled by bolt with respect to elevator is $S_{BE} = 1.2\ \text{m}$
$\frac{1}{2} \times a_{BE} \times t^2 = S_{BE}$
$t = \sqrt{\frac{2\times1.2}{12}} = \frac{1}{\sqrt{5}}s$

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CHEMISTRY

(46) Answer : (3)
Solution:
For ionic solid, formula mass is used instead of molecular mass.

(47) Answer : (3)
Hint:
Mass of 1 atom of He = 4 u
Mass of 4 atoms of He = 4 × 4 = 16 u

(48) Answer : (2)
Solution:
Empirical formula is the simplest whole number ratio of elements present in a compound.

Element	%	Mole	Ratio
C	40	$\frac{40}{12} = 3.33$	$\frac{3.33}{3.33} = 1$
H	6.67	$\frac{6.67}{1} = 6.67$	$\frac{6.67}{3.33} = 2$
O	53.3	$\frac{53.3}{16} = 3.33$	$\frac{3.33}{3.33} = 1$

Empirical formula is CH_2O

(49) Answer : (3)

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Solution:
Precision refers to the closeness of various measurements for the same quantity.
Accuracy refers to agreement of a particular value to the true value of the result.

(50) **Answer :** (3)
Solution:
Moles of CH_4 = $\frac{1.6}{16}$ = 0.1 mol
Number of electrons of CH_4 = 0.1 × 10 × N_A
= 6×10^{23}

(51) **Answer :** (2)
Hint:
Concentration terms involving volume are temperature dependent.

(52) **Answer :** (4)
Solution:
Pure substance have a fixed composition. Example: Carbon dioxide

(53) **Answer :** (4)
Hint:
One molecule of Biomolecule should contain one atom of Zn.
Solution:
$\% \text{ of Zn} = \frac{\text{Number of Zn atom} \times 65.38}{\text{wt. of biomolecule}} \times 100$
$0.75 = \frac{1 \times 65.38}{MW} \times 100$
∴ Minimum molar mass of biomolecule
$= \frac{65.38 \times 100}{0.75}$
= 8717.33 g/mol

(54) **Answer :** (3)
Solution:
Let the percentage abundances of two isotopes are x and (100 – x) respectively.

$12.2 = \frac{(x \times 12) + (100 - x) \times 13}{100}$
$\Rightarrow$ 1220 = 12x + 1300 – 13x
$\Rightarrow$ x = 80%
∴ Percentage abundance of 2nd isotope
= 100 – 80 = 20%

(55) **Answer :** (2)
Solution:
Mass of 1 mol (6.022×10^{23} atoms) of H_2O molecules = 18 g
∴ Mass of 1 H_2O molecule = g $\frac{18}{6.022 \times 10^{23}} \approx 3 \times 10^{-23}$

(56) **Answer :** (4)
Hint:
Number of atoms = Moles × N_A × Atomicity
Solution:
Let mass of each of S_2 and S_8 taken be x g

S_2	S_8
Moles = $\frac{x}{32 \times 2}$	$\frac{x}{32 \times 8}$
Molecules = $\frac{x}{32 \times 2} \times N_A$	$\frac{x}{32 \times 8} \times N_A$

Let mass of each O_2 and O_3 taken be x g
Number of atoms in O_2 molecule $= \frac{x}{16 \times 2} N_A \times 2 = \frac{x}{16} N_A$
Number of atoms in O_3 molecule $= \frac{x}{16 \times 3} N_A \times 3 = \frac{x}{16} N_A$

(57) **Answer :** (4)
Solution:

Number	Significant figures

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2.005	4
1.32×10^2	3
4.9500	5
0.0024	2

(58) **Answer :** (1)
Solution:

Moles of acetic acid = $\frac{6.022 \times 10^{20}}{6.022 \times 10^{23}}$ = 10^{-3} moles

Molarity = $\frac{10^{-3}}{500} \times 1000 = 2 \times 10^{-3}\mathrm{M} = 0.002\ \mathrm{M}$

(59) **Answer :** (1)
Solution:
$1 \times 500 = 0.2 \times V_2$
$V_2 = 2500$
∴ Volume of water added
= 2500 – 500 = 2000 mL = 2 L

(60) **Answer :** (3)
Solution:
1 mole of NO_2 will occupy 22.4 L at STP.

Density $= \frac{\text{Molar mass}}{\text{Molar volume}} = \frac{46}{22.4} = 2.05$ g/L

(61) **Answer :** (4)
Solution:
$M_1V_1 + M_2V_2 = M_3V_3$.
$100 \times 5 + 800 \times 2.5 = M_3 \times (100 + 800)$

$M_3 = \frac{500+2000}{900} = 2.78$ M

(62) **Answer :** (1)
Solution:
In 1 kg of 0.1 molal glucose solution, mass of glucose = 180 g
∴ Mass of solution = 1000 + 180 = 1180
⇒ 1180 g of glucose solution ≡ 180 g glucose
∴ 2000 g of glucose solution $= \frac{180}{1180} \times 2000 \approx 305g$

(63) **Answer :** (3)
Solution:
$3BaCl_2 + 2Na_3PO_4 \rightarrow Ba_3(PO_4)_2 + 6NaCl$
0.3 mole of $BaCl_2$ combines with 0.2 mole of Na_3PO_4. Therefore, $BaCl_2$ is limiting reactant.
0.3 mole of $BaCl_2$ will produce 0.1 mole of $Ba_3(PO_4)_2$ and 0.6 mole of NaCl.

(64) **Answer :** (2)
Solution:

$$\underset{2\ \text{Volume}}{2H_2\,(g)} + \underset{1\ \text{Volume}}{O_2\,(g)} \rightarrow 2H_2O$$

(65) **Answer :** (3)
Solution:
In $(NH_4)_3PO_4$ 12 moles of 'H' are present with 4 moles of oxygen atom.
∴ 3.18 moles of 'H' are present with = $\frac{4}{12} \times 3.18$ = 1.06 moles of oxygen atom.

(66) **Answer :** (2)
Hint:
$CaCO_3$ decomposes on heating into CaO and CO_2.
Solution:
$CaCO_3(s) \rightleftharpoons CaO(s) + CO_2(g)$
1 mol that is 100 g $CaCO_3$ gives → 22.4 L CO_2
Volume of CO_2 = $\frac{50}{100} \times \frac{25}{100} \times 22.4 = 2.8\ \mathrm{L}$

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(67) **Answer :** (3)
Solution:
100 g solution would have 49 g H_2SO_4

Number of mole of $H_2SO_4 = \frac{49}{98} = 0.5$

Density of solution $= \frac{\text{Mass of solution}}{\text{Volume of solution}}$

$2 = \frac{100}{V(mL)}$

$\text{Molarity} = \frac{0.5}{50} \times 1000$

= 10 mol/l

(68) **Answer :** (4)
Hint:
Atoms are conserved in chemical reactions.

(69) **Answer :** (2)
Solution:
As per the law of multiple proportions, if two elements combine to form more than one compound the masses of one element that combine with a fixed mass of the other element, are in the ratio of small whole numbers.

(70) **Answer :** (3)
Solution:
- Number of Mg atoms = $\frac{1}{24} \times N_A$
- Number of O atoms = $\frac{1}{32} \times 2 \times N_A$
- Number of Li atoms = $\frac{1}{7} \times N_A$
- Number of Ag atoms = $\frac{1}{108} \times N_A$

(71) **Answer :** (3)
Solution:
Urea (NH_2CONH_2):
% of C = $\frac{12}{60} \times 100$ = 20%
Glucose ($C_6H_{12}O_6$):
% of C = $\frac{12 \times 6}{180} \times 100$ = 40%
Methane (CH_4):
% of C = $\frac{12}{16} \times 100$ = 75%
Carbon dioxide
% of C = $\frac{12}{44} \times 100$ = 27.2%

(72) **Answer :** (1)
Hint:
'H_2' is the limiting reactant.
Solution:

$N_2 + 3H_2 \rightleftharpoons 2NH_2$

1 3/2

Number of moles of NH_3 formed is decided only by H_2. Number of moles of NH_3 formed
= $\frac{2}{3} \times \frac{3}{2}$ = 1
1 mole NH_3(g) formed
i.e. 22.4 L NH_3(g) formed

(73) **Answer :** (1)
Hint:
Left moles of CO_2 can be calculated as
mole left = Initial mole – Removal moles
Solution:
Remaining molecules of CO_2 = $\left(\frac{220}{44} \times 10^{-3} \times 6 \times 10^{23}\right) - 3 \times 10^{20}$
= $(3 \times 10^{21}) - (3 \times 10^{20})$
= 27×10^{20}

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= 2.7 ×10^{21}

Remaining atoms of CO_2 = 2.7 × 10^{21} × 3

= 8.1 × 10^{21}

(74) Answer : (1)
Solution:
a. 18 g $C_6H_{12}O_6$
Number of oxygen atoms
= $\frac{18}{180} \times 6 \times N_A$
= 0.6 N_A
b. 44.8 L NO_2 gas at STP
Number of oxygen atoms
= $\frac{44.8}{22.4} \times 2 \times N_A = 4\ N_A$
c. 1.225 g $KClO_3$
Number of oxygen atoms
$= \frac{1.225}{122.5} \times 3 \times N_A = 0.03\ N_A$
d. 11.2 L CO gas at STP
Number of oxygen atoms
$= \frac{11.2}{22.4} \times 1 \times N_A = 0.5\ N_A$

(75) Answer : (1)
Solution:
$Na_2SO_4 \rightarrow 2Na^+$
$0.05 \times \frac{250}{1000}$ mole $= 0.0125$ moles of Na_2SO_4
2 × 0.0125 = 0.0250 = 2.5 × 10^{-2} moles of Na^+

(76) Answer : (3)
Solution:
2A + B = 80 …(i)
2A + 3B = 100 …(ii)
From equations (i) and (ii)
A = 35 u and B = 10 u

(77) Answer : (3)
Solution:
Mass of 1 atom = 1.8 × 10^{-22} g
Mass of 6.022 ×10^{23} atoms
= 1.8 × 10^{-22} × 6.022 ×10^{23} = 108.39 g
= Atomic mass

(78) Answer : (1)
Hint:
$Molality\ (m) = \frac{Mole\ of\ solute}{wt.\ of\ solvent\ (kg)}$
Solution:
n_{urea} = 1
$W_{H_2O} = 1\ kg = 1000g$
$X_{urea} = \frac{n}{n+N} = \frac{1}{1+\frac{1000}{18}} = \frac{1}{56.55} = 0.018$

(79) Answer : (3)
Hint:
Mole of $CO_2 = \frac{\text{Volume of } CO_2 \text{ at STP}}{\text{Molar volume at STP}}$
Solution:
Mole of $CO_2 = \frac{224}{22400}$
Mole of O atoms $= \frac{224}{22400} \times 2 = 2 \times 10^{-2}$
Number of O atoms = 2 × 10^{-2} × N_A

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(80) **Answer :** (2)
Solution:
Let the temperature be x

$^{\circ}\mathrm{F} = \frac{9}{5}(^{\circ}\mathrm{C}) + 32$

$\mathrm{x} = \frac{9}{5}\mathrm{x} + 32$

$-\frac{4\mathrm{x}}{5} = 32$

x = –40

(81) **Answer :** (3)
Solution:

	$CO_2(g)$	+ C(s)	⟶	2CO(g)
Initial	1L	0		0
Final volume	(1 – x)			2x

Final volume = 1 – x + 2x = 1.4 L

∴ $\boxed{\mathrm{x} = 0.4\mathrm{L}}$

∴ Volume of CO = 2x = 2 × 0.4 = 0.8 L
Volume of CO_2 = (1 – x) = 1 – 0.4 = 0.6 L

(82) **Answer :** (2)
Solution:
The SI unit for amount of substance is mole.
The SI unit for mass is kg.

(83) **Answer :** (1)
Solution:
P_4 + 5O→ P_4O_{10}
31g
$\frac{1}{4}$ mol P_4 gives $\frac{1}{4}$ mol P_4O_{10} for 100% yield or $\frac{1}{4} \times \frac{60}{100}$ mol P_4O_{10} when 60% yield
Mol of $P_4O_{10} = \frac{1}{4} \times \frac{60}{100}$ mol
Mass of P_4O_{10} = n × M_O = $\frac{1}{4} \times \frac{60}{100} \times 284 = 42.6\,g$
(because MW (P_4O_{10}) = (31 × 4) + (16 ×10) = 284)

(84) **Answer :** (4)
Hint:
Multiple for giga is 10^9
Solution:

Prefix	Multiple
Mega	10^6
Giga	10^9
Exa	10^{18}
Tera	10^{12}

(85) **Answer :** (3)
Solution:

Compound	Molecular formula	Empirical formula
Benzene	C_6H_6	CH
Ethyne	C_2H_2	CH
Cyclohexane	C_6H_{12}	CH_2
Propyne	C_3H_4	C_3H_4

(86) **Answer :** (2)
Hint:

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$\underset{(s)}{Mg} + \underset{(aq.)}{H_2SO_4} \rightarrow \underset{(aq.)}{MgSO_4} + \underset{(g)}{H_2}$

Solution:
Mole of Mg = mole of H_2

$\frac{wt.}{24} = \frac{112}{22400}$

wt. of Mg = 0.12 g

(87) **Answer :** (2)
Solution:
Let the mass of NaOH dissolved = xg

$\therefore$ Mass% = 20 = $\frac{x}{(60+x)} \times 100$

$\Rightarrow$ 1200 + 20 x = 100 x

$\therefore x = \frac{1200}{80} \Rightarrow 15$ g

(88) **Answer :** (2)
Solution:

$n = \frac{1.5055 \times 10^{23}}{6.022 \times 10^{23}} = 0.25$ mol

Mass of 1 mole of CO = 28 g
Mass of 0.25 mol of CO = 7 g

(89) **Answer :** (4)
Solution:
1 mol $N_2 \equiv$ 2 mol NO $\equiv$ 2 mol NO_2
$\therefore$ 0.5 mol $N_2 \equiv$ 1 mol NO $\equiv$ 1 mol NO_2
Mass of NO_2 = 1 × 46 = 46 g

(90) **Answer :** (3)
Hint:

$\text{Molarity} = \frac{n_{\text{solute}}}{\text{Volume of solution (in L)}}$

Solution:
Moles of HCl = $\frac{0.01 \times 500}{1000} \Rightarrow 0.005$
1 mol of HCl requires 1 mole of NaOH as per given equation.
Hence mass of NaOH = 0.005 × 40 = 0.2 g

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BOTANY

(91) **Answer :** (3)
Hint:
Lipid molecules of plasma membrane have polar head and non-polar tail.
Solution:
Polar head is hydrophilic in nature and it interacts with water. Rest all the statements regarding plasma membranes are true.

(92) **Answer :** (3)
Solution:
Small hair like structures, which work like oars causing the movement of either the cell or surrounding fluid is cilia.

(93) **Answer :** (2)
Solution:
Chromosomes having one shorter arm and one longer arm exhibit L-shape during anaphase. Such chromosomes are called sub-metacentric.

(94) **Answer :** (2)
Hint:
Hub in centriole is proteinaceous.
Solution:
The central part of the proximal region of the centriole is proteinaceous and called the hub.

(95) **Answer :** (3)
Solution:

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Chloroplasts and mitochondria are semi-autonomous cell organelles.
Mitochondria, chloroplast and bacteria, all have double stranded circular DNA, 70S ribosomes and has the ability to self duplicate.

(96) **Answer :** (3)
Solution:
In plants, Golgi apparatus is made up of unconnected units called dictyosome.

(97) **Answer :** (2)
Solution:
Chromoplasts are coloured plastids. They are formed either from leucoplasts or chloroplasts.

(98) **Answer :** (1)
Solution:
Microbodies are present in both animals and plants.

(99) **Answer :** (2)
Solution:
Eukaryotic flagella typically have a "9 + 2" arrangement of microtubules.
In eukaryotic flagella, there are 20 microtubules.

(100) **Answer :** (4)
Solution:
Chromatin contains DNA and some basic proteins called histones. The cell wall and middle lamellae may traversed by plasmodesmata which connect the cytoplasm of neighbouring cells.

(101) **Answer :** (3)
Solution:
Lysosomes contain hydrolytic enzymes which become active at acidic pH.
If pH in lysosomes is increased to alkaline or basic then hydrolytic enzymes will become inactive.

(102) **Answer :** (3)
Hint:
The plasmid DNA confers certain unique phenotypic characters, such as resistance to antibiotics, to some bacteria.
Solution:
Bacteria are prokaryotes and prokaryotes have double stranded circular DNA as genetic material.

(103) **Answer :** (4)
Hint:
Active transport is an energy dependent process of cell membrane.
Solution:
Cell wall protects the cells from bursting, functions as barrier to undesirable macromolecules and also protects the cell from infections.

(104) **Answer :** (4)
Solution:
The shape of mesophyll cell is round and oval.

(105) **Answer :** (3)
Solution:
Rudolf Virchow explained that cells divide and new cells are formed from the pre-existing cells (*Omnis cellula-e cellula*).

(106) **Answer :** (4)
Hint:
Schwann proposed hypothesis that the bodies of animals and plants are composed of cell and product of cell.
Solution:
Theodore Schwann studied different type of animal cells and reported that cell had a thin outer layer.

(107) **Answer :** (3)
Solution:
Thick and tough glycocalyx is called capsule. It is present in some bacteria.

(108) **Answer :** (2)
Hint:
The size of a typical bacterium is generally 3-5 µm.
Solution:
Mycoplasma, the smallest cell, is only 0.3 µm long and it lacks a nuclear membrane.

(109) **Answer :** (1)
Solution:

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Golgi apparatus was first observed and described by Camillo Golgi. Camillo Golgi described Golgi apparatus as a densely stained reticular structures present near the nucleus of the cell. Therefore, these were given the name Golgi body, after his name.

(110) Answer : (1)
Solution:
Cytoplasm is the main arena of cellular activities. It is the semi-fluid matrix and contains many cell organelles.

(111) Answer : (3)
Solution:
In some prokaryotes like cyanobacteria, there are other membranous extensions into the cytoplasm called chromatophores which contain pigments.

(112) Answer : (2)
Solution:
Leeuwenhoek was the first to observe a living cell.

(113) Answer : (4)
Solution:
The membrane of RBCs in human beings is composed of approx. 52% protein and 40% lipids.

(114) Answer : (4)
Hint:
Kinetochore is the site where microtubules of spindle attach.
Solution:
On the side of centromere of chromosome, disc shaped structures are present known as kinetochore.

(115) Answer : (4)
Hint:
Some of the organelles function in a co-ordinated manner and constitute an endomembrane system.
Solution:
The organelles included in the endomembrane system are Endoplasmic reticulum, Golgi complex, Lysosomes and Vacuoles.

(116) Answer : (2)
Solution:
At number of places the nuclear envelope is interrupted by minute pores, which are formed by the fusion of its two membranes.

(117) Answer : (3)
Solution:
Bacterial filament is composed of three parts – filament, hook and basal body.

(118) Answer : (2)
Solution:
Centrioles are absent in higher plants.

(119) Answer : (2)
Hint:
Secondary constriction is a non-staining region of SAT chromosome.
Solution:
Satellite is a region beyond secondary constriction in SAT chromosome.

(120) Answer : (2)
Solution:
Stroma lamellae are flat membranous tubules connecting different grana.
Stroma is the space limited by inner membrane of the chloroplast.
Thylakoids are flattened membranous sacs present in the stroma of chloroplast.
Grana are the thylakoids arranged in stacks.

(121) Answer : (4)
Solution:
Viruses are exception to the cell theory as they are acellular.

(122) Answer : (2)
Hint:
Chitin is present in the cell wall of fungi.
Solution:
Algae have cell wall made of mannans, cellulose, galactans and minerals like calcium carbonate.

(123) Answer : (3)

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Hint:
Smooth endoplasmic reticulum (SER) is free of ribosome.
Solution:
Detoxification of drugs, muscle contraction, synthesis of steroids and lipids are the function of SER. Cell wall helps in cell to cell interaction.

(124) Answer : (4)
Solution:
In the given diagram,
A – Central sheath
B – Central microtubule doublet
C – Radial spoke
D – Plasma membrane

(125) Answer : (3)
Solution:
Microtubules help in cell plate formation during plants cytokinesis.
Formation of cleavage furrow during animal cell division occur with the help of microfilaments.

(126) Answer : (3)
Hint:
Inclusion bodies include phosphate granules, cyanophycean granules and glycogen granules.
Solution:
Contractile vacuole is an example of vacuole and it is a part of endomembrane system.

(127) Answer : (3)
Solution:
Gas vacuoles are found in blue-green and purple and green photosynthetic bacteria.

(128) Answer : (2)
Solution:
Golgi complex is involved in glycosylation of proteins.
Contractile vacuole – Excretion
RER – Protein synthesis
Amyloplast – Stores starch

(129) Answer : (3)
Solution:
Telomere does not represent primary constriction. Telomeres seal the ends of chromosomes and prevent their shortening or chromosome loss.

(130) Answer : (1)
Solution:
Mature sieve tube cell lacks nucleus.

(131) Answer : (2)
Solution:
Loose and indistinct network of nucleoprotein fibres present in interphase nucleus is called chromatin.

(132) Answer : (1)
Solution:
Human cheek cell does not have cell wall, but cell of onion peel has cell wall.

(133) Answer : (2)
Solution:
Electron microscopic studies of eukaryotic cells reveal the presence of a network or reticulum of tiny tubular structures scattered in the cytoplasm that is called ER. Hence. Endoplasmic reticulum divides the intracellular space into two distinct compartments.

(134) Answer : (3)
Hint:
Fungal cell wall is made up of polymer of N-acetylglucosamine units.
Solution:
Primary cell wall of plant, gradually diminishes with maturity.

(135) Answer : (1)
Solution:
Integral proteins are partially or totally buried in the membrane.

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ZOOLOGY

(136) Answer : (1)
Solution:
Smooth muscle fibres are uninucleated and fusiform in shape.

(137) Answer : (4)
Solution:
Neuroglial cells support and protect neurons.

(138) Answer : (1)
Solution:
Goblet cells of alimentary canal are modified from columnar epithelial cells which secrete mucus.

(139) Answer : (4)
Hint:
Its cells are formed in bone marrow.
Solution:
Collagen or elastin fibres are absent in fluid connective tissue (*e.g.,* blood).

(140) Answer : (4)
Solution:
Skeletal connective tissue like bone and cartilage are supportive connective tissues.

(141) Answer : (4)
Solution:
Squamous epithelium forms the innermost layer of walls of blood vessels (endothelium), air sac (alveoli) of the lungs.

(142) Answer : (2)
Solution:
Adipose tissue is concerned with storage of fat.
Structural framework is provided by areolar tissue.

(143) Answer : (4)
Solution:
Smooth muscles have unstriated, uninucleated, and fusiform-shaped muscle fibres. They are present in the walls of blood vessels, stomach, intestine, *etc.,* conversely, the heart has cardiac muscles, which are characterised by striated, cylindrical, branched, and uninucleated muscle fibres.

(144) Answer : (4)
Solution:
The intercalated discs function as boosters of the contraction wave, permitting the wave of muscle contraction to be transmitted from one cardiac fibre to another. Thus, all the fibres contract as a single unit.

(145) Answer : (3)
Solution:
Bronchioles and oviducts are lined by simple ciliated columnar epithelium. Simple squamous epithelium is present in alveoli of lungs.

(146) Answer : (3)
Solution:
Cell junctions are also present in simple epithelial tissue, muscular tissue, *etc*.

(147) Answer : (2)
Solution:
Dendrites are afferent processes whereas the axon is the efferent process of a neuron.

(148) Answer : (1)
Solution:
Neural tissue exerts the greatest control over the body's responsiveness to changing environmental conditions.

(149) Answer : (1)
Solution:
Fibres and fibroblasts are compactly packed in the dense connective tissues. Orientation of fibres show a regular or irregular pattern and are called dense regular and dense irregular connective tissues respectively.

(150) Answer : (3)

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Solution:
The columnar epithelium composed of tall and slender cells with basal nuclei is found in the lining of stomach and intestine and helps in secretion and absorption.

(151) Answer : (2)
Solution:
Shortening and lengthening of the muscle fibres occur due to the property of contractility.
Nerve fibres do not show contractility. Insulation in nerve fibres is provided by myelin sheath.

(152) Answer : (3)
Solution:
Epithelial tissue is avascular. Simple epithelial cells form a single layer. Compound epithelium lines the urinary bladder and ureters.

(153) Answer : (1)
Solution:
The only unicellular gland is goblet cell and rest of the glands are multicellular.

(154) Answer : (1)
Hint:
It serves as a support framework for epithelium.
Solution:
Areolar tissue is an example of loose connective tissue
Dermis of skin possesses dense, irregular connective tissue
Blood and bone are specialised connective tissues.

(155) Answer : (4)
Solution:
Tendons connect skeletal muscles to bones. They are categorised under dense regular connective tissue.
Presence of lamellae is the characteristic of bones.

(156) Answer : (4)
Hint:
Blood is a fluid connective tissue.
Solution:
In all connective tissues except blood, the cells secrete fibres of structural proteins called collagen or elastin. The fibres provide strength, elasticity and flexibility to the tissue. These cells also secrete modified polysaccharides.
Areolar tissue contains fibroblasts, macrophages and mast cells.

(157) Answer : (1)
Hint:
Muscle fibres associated with limbs and heart are striped in appearance.
Solution:
Striated muscles include both cardiac and skeletal muscles. The alternate light and dark bands refer to the horizontal arrangement of actin and myosin filaments.

(158) Answer : (3)
Hint:
Skeletal muscle fibres show syncytium.
Solution:

Muscle fibres of biceps	**Muscle fibres of heart**
Voluntary	– Involuntary
Unbranched	– Branched
Multinucleated	– Uninucleated
Striated	– Striated

(159) Answer : (2)
Hint:
Identify a polymer of fructose.
Solution:
Inulin is obtained from the roots of plants like *Dahlia*. It is a polymer of fructose whereas starch, glycogen and cellulose are polymers of glucose.

(160) Answer : (1)
Hint:
Each glycosidic bond is formed by formation of one water molecule.
Solution:

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Both X and Y are hexose sugars which join to form a disaccharide (Z) by formation and release of one water molecule. So, reaction will be

$$C_6H_{12}O_6 + C_6H_{12}O_6 \xrightarrow{-H_2O} C_{12}H_{22}O_{11}$$

(161) Answer : (3)
Solution:
In a polysaccharide chain (like glycogen), the right end is called the reducing end and the left end is called the non-reducing end. The reducing end of a sugar has free aldehydic group, when its ring structure opens.

(162) Answer : (2)
Solution:
Living tissue forms thick slurry when grind with trichloroacetic acid. When slurry is strained through cheesecloth, yields two fractions: Acid-soluble pool (micromolecules) and acid-insoluble fraction (macromolecules).

(163) Answer : (3)
Hint:
Homopolymer of glucose
Solution:
Cellulose is a homopolysaccharide consisting of only one type of monosaccharide units *i.e.,* glucose. Unlike starch, cellulose does not contain complex helices and hence, cannot hold I_2 molecules. Since it cannot hold I_2, it remains colourless during the test.

(164) Answer : (4)
Solution:
Abrin and ricin are toxins. Concanavalin A is a lectin.

(165) Answer : (2)
Solution:
Silicon is present in negligible amount in human body.

(166) Answer : (3)
Solution:
Primary metabolites such as amino acids and sugars have identifiable functions in normal physiological processes.

(167) Answer : (2)
Solution:
A polysaccharide made of 458 glucose molecules have formed by 457 glycosidic bonds. When completely hydrolyzed, each glycosidic bond is broken by using one water molecule. So 457 water molecules will be consumed in the reaction.

(168) Answer : (3)
Hint:
This sugar is abundant in corn and grapes.
Solution:
Fructose is six carbon sugar with keto group. Ribose and deoxyribose are five carbon sugars with aldehyde group.

(169) Answer : (4)
Solution:

Component	% of the total cellular mass
Water	70-90
Proteins	10-15
Carbohydrates	3
Lipids	2
Nucleic acids	5-7
Ions	1

(170) Answer : (3)
Solution:
In chitin, the basic unit is a nitrogen containing glucose derivative known as N-acetylglucosamine.

(171) Answer : (1)
Solution:
Macromolecules are present in the acid-insoluble fraction *e.g.*, polysaccharides, proteins and nucleic acids.
Micromolecules are present in the acid-soluble fraction *e.g.,* amino acids, simple sugars and nucleotides.

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(172) Answer : (4)

Solution:

Both starch and cellulose are found in the retentate obtained upon the chemical analysis of a living tissue.
The most abundant carbohydrate on Earth is cellulose. It is a complex polysaccharide found in the cell walls of plants and is the primary structural component of wood and cotton.
Starch is a storage homopolysaccharide whereas cellulose is a structural homopolysaccharide. Starch contains complex helices in its structure.

(173) Answer : (2)

Solution:

While primary metabolites have identifiable functions and play known roles in normal physiological processes. The role or functions of all the 'secondary metabolites' in host organisms is still not understood at the moment. However, many of them are useful to 'human welfare' (*e.g.*, rubber, drugs, spices, scents and pigments). Some secondary metabolites have ecological importance.

(174) Answer : (1)

Solution:

Exoskeletons of arthropods have a complex homopolysaccharide called chitin. It's monomeric unit is N-acetyl glucosamine [NAG].

(175) Answer : (2)

Solution:

Nucleic acids serve as genetic material and constitute 5-7% of the total cellular mass.

(176) Answer : (2)

Hint:

The cells that secrete collagen or elastin are absent in blood.

Solution:

In all connective tissues, except blood, the cells secrete fibres of structural proteins called collagen or elastin. The fibres provide strength, elasticity and flexibility to the tissues.

(177) Answer : (4)

Solution:

Stratified epithelium has more than one layer. Due to the presence of mucus secreting goblet cells, columnar epithelium is also known as simple columnar glandular epithelium.

(178) Answer : (1)

Solution:

Glycogen and starch are homopolymers of glucose. Starch is a storage polysaccharide is case of plants while glycogen is a storage polysaccharide in case of animals.

(179) Answer : (4)

Solution:

Cellulose is the most abundant structural polysaccharide in nature and is also a secondary metabolite categorised as a polymeric substance.

(180) Answer : (2)

Solution:

Element	% weight of human body
Hydrogen	9.5
Carbon	18.5
Oxygen	65.0
Calcium	1.5

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